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Metamer Mismatch Bodies: Foundations, Methods, and Applications in Color Science

Alexander Forsythe^{1,2}, Brian Funt¹

¹Simon Fraser University, Vancouver, Canada

²Academy of Motion Picture Arts and Sciences, Los Angeles, United States

E-mail: agf8@sfu.ca

Abstract. Metameric object matching is an intrinsic characteristic of trichromatic color measurement where spectrally distinct object reflectances produce identical color signals under a given viewing condition. However, when the viewing conditions change, the previously identical color signals may diverge in a phenomenon known as metamer mismatching. In this paper we review the conceptual foundations, evolving computational methods, and practical applications of Metamer Mismatch Bodies (MMBs), which characterize the range of color signals produced by a metamer set with a change in viewing conditions. We outline advancements in models of metamer mismatching from early statistical estimates and linear programming to modern algorithms capable of computing precise mismatch boundaries without assumptions about reflectance smoothness or transition count. We review the use of MMBs in practice for light source design and evaluation, digital camera sensor design and color appearance modeling. And finally, we present an optimized MATLAB implementation of the Logvinenko et al. five-transition approximation algorithm enabling large-scale spectral analysis and broader integration into imaging pipelines. By consolidating theoretical developments and practical advances, this survey positions MMBs as a foundational tool for understanding and quantifying color variation across changing conditions.

1 Introduction

Metamerism is a phenomenon in which two objects with different spectral reflectance functions produce the same color signal (i.e., triple of sensor responses) for a given set of conditions (e.g., light source and observer) [1, 2, 3]. Such spectrally different objects are referred to as metamers [2]. Color imaging systems, such as the human visual system and most common color imaging systems, reduce high-dimensional spectral information into three signals (e.g., CIE XYZ, LMS, RGB, etc.), resulting in the loss of potentially useful details about the reflectance of objects [2]. This highlights the inherent limitations of trichromatic color representation. For any given color signal under a specific set of conditions, there exists an infinite set of spectrally different object reflectances that are metameric to it. This collection is known as the metamer set [2, 3].

This inherent loss of spectral information leads to the phenomenon of metamer mismatching. [4, 3]. Metamer mismatching describes the breakdown that results from this reduction when conditions vary, causing the color signals which were identical under one set of conditions (the reference light source and observer) to no longer be the same when viewed under a different set of conditions (the test light source and observer) [4, 3]. As a consequence, two objects appearing the same under one light source can be expected to appear different under a second light source [4, 3]. The resulting color differences can be significant, potentially leading to large color shifts [4, 5, 3].



One potentially useful way of quantifying and visualizing the effect of metamerism and the resulting metameric mismatching is through Metamer Mismatch Bodies (MMBs) [5, 4, 3, 6, 7]. An MMB defines the boundaries in color space of the set of all possible color signals that can be produced by the members of a given metamer set when viewed under a new set of conditions [5, 4, 3, 6, 7]. These volumes represent the range of distinguishable signals arising from objects that were metameric under the original conditions [4, 3]. Understanding the extent of metamer mismatching often requires determining the precise boundary of the MMB in color space [4, 3, 3], a challenge that has been a focus of research.

The study of metamerism, including its quantitative assessment and the potential for mismatch, has a long history in color science [5, 3]. Early research explored methods for evaluating metameric colors [8, 9], quantifying the degree of metamerism [10, 9], and counting metameric object colors under different conditions [10, 11]. Foundational work, such as studies by MacAdam [12] and Stiles and Wyszecki [10], investigated the theoretical limits of object colors. These limits are often described in terms of optimal colors [13, 14, 15]. Optimal are those lying on the boundary of the object-color solid [13, 14]. Optimal colors for human vision were historically thought to result from optimal reflectance spectra that take on only the values 0 and 1 [13, 16, 17, 12, 15] with at most two transitions between those values [13, 16, 17, 12, 15, 18, 19]. However, this two-transition assumption has been shown to be partially incorrect for human spectral sensitivity functions [13, 16, 17, 18, 14]. A more recent algorithm accurately computes the true object-color solid boundary [14], providing a theoretical framework to establish its true shape, which was previously unknown. Logvinenko et. al [14] prove that optimal reflectances for human vision include those with up to four transitions [14, 17, 18, 13], showing that Schrödinger's optimal color theorem [19] does not strictly generalize in the way previously thought [13, 16, 14, 18]. Consequently, this work provides a more precise definition of the true object-color solid boundary than previous methods based solely on the two-transition approximation [13, 14, 18].

Precisely computing the exact boundary of the MMB for any possible reflectance spectrum, without placing restrictions on the spectral reflectance functions, remained an unsolved problem for a long time [3]. Recent advancements have provided algorithms capable of precisely characterizing this boundary [3, 20]. Although their implementation provides only approximations based on a simplifying assumption, the theory introduced by Logvinenko, Funt, and Godau provides the theoretical background for computing the precise MMB boundary for arbitrary, strictly positive illuminants and sensor functions [3, 21], marking a significant step towards a complete understanding of the potential extent of metamer mismatching [4, 3]. The development of such precise computational methods paves the way for exploring the full implications of metamer mismatching in various applications.

2 Metamer Mismatch Body Calculation Algorithms

Calculating Metamer Mismatch Bodies (MMBs), also known as Metamer Mismatch Volumes (MMVs), Metameric Mismatch Gamuts, and Color Mismatch Volumes (CMV), has been the subject of ongoing study in color science for decades [5, 8, 10, 7, 22, 3, 23, 20]. These volumes describe the range of color signals that metameric surface reflectance functions may produce under a different set of conditions [24, 15, 2, 3, 23, 20]. The phenomenon of metamer mismatching reveals fundamental limitations of trichromatic color representation [24, 4, 3, 25] and has significant implications for color appearance, reproduction, and visual perception [24, 15, 4, 3, 20].

Early approaches to MMB estimation relied on statistical or heuristic methods to approximate the boundaries of these volumes [15, 9, 3], often using simplified models of surface reflectance and vision [15, 5, 9]. Later work introduced more rigorous and computationally intensive techniques aimed at precisely characterizing these boundaries [3, 25, 20]. Over time, the focus shifted from probabilistic representations [15, 9] to exact geometric descriptions [20, 3, 23] rooted in optimization and linear programming [15, 22, 23].

This review surveys the evolution of MMB computation methods, tracing the progression from early statistical estimates to contemporary boundary-solving algorithms. It examines both theoretical developments and practical implementations, highlighting the assumptions, strengths, and limitations of each major approach.

Early Statistical Methods

Research into MMBs began with techniques intended to estimate the set of possible color signals that could arise from a change in light source or sensing device. Early methods primarily employed statistical techniques [10, 8, 9, 15, 5] or simplified theoretical models [22, 26, 27, 7, 15, 28, 29, 3] to approximate these volumes [10, 15, 22, 27, 5, 3, 25, 4].

Metameric Black and Monte Carlo Methods

One foundational technique involved the concept of metameric blacks [8, 1]. A metameric black is a spectral reflectance function that produces zero tristimulus values under a given illuminant and observer [8, 1]. Wyszecki, in 1958, described a method based on precalculated sets of linearly independent metameric blacks that could be linearly combined with an initial spectral reflectance function (which itself produces the target tristimulus values) to generate any number of metameric colors [8, 15]. This approach allowed for the generation of a large number of metamers, making it well-suited for computational studies [8, 9]. The method included a scaling factor to ensure resulting reflectances stayed within the physical range of 0 to 1 [8, 15].

Building on this, Stiles and Wyszecki applied elementary statistical methods to the problem of “counting metamers” [10] or determining the proportion of object colors from a collection that are approximately metameric to a given color. This statistical procedure aimed to quantify the intuitive idea that the number of metamers increases as one moves inward from the boundary of the object color solid [10], an idea later shown to hold both in theory and in practice [24]. They introduced the concept of “basic collections” of spectral reflectance functions, including simplified ones like step functions, while acknowledging that such assumptions (like independence of reflectance values at different wavelengths) were often not realistic for practical materials [10]. They demonstrated how the distribution of stimulus values could be approximated by a Gaussian distribution for large numbers of wavelength intervals, based on the Central Limit Theorem [10]. However, they noted that this normal approximation breaks down away from the center of the distribution or near the boundary of the object-color space [10, 11]. The metameric black method was seen as a Monte Carlo approach that could be used when the assumption of normality failed [10]. Comparisons showed that statistical boundaries (ellipsoids) derived from the deterministic method (based on normality) and the Monte Carlo metameric black method showed good agreement in shape and orientation, although their absolute volumes could differ significantly [10].

Takahama and Nayatani [30] later proposed a method for generating metameric stimuli by adjusting initial spectral reflectance functions to yield target tristimulus values, offering a simpler computational procedure compared to prior methods [30]. This technique was particularly notable for its ability to produce smoothed metameric spectral-reflectance functions that more closely resembled actual paint and dye products, and could also be extended to generate metamers under multiple illuminant-observer combinations simultaneously [30].

Deterministic and Statistical Boundary Estimation

Stiles and Wyszecki’s deterministic approach aimed to predict statistical boundary ellipsoids without requiring the actual generation of spectral reflectance functions [10]. This involved assuming a basic collection of spectral reflectance functions (initially “jagged” step functions) and deriving the distribution of tristimulus values, assuming normality [10]. They later extended this by studying basic collections of frequency-limited spectral reflectance functions which could be made smoother and more representative of real objects [11]. Surprisingly, they found that the 95%-boundary ellipsoid calculated using smooth, frequency-limited functions was very similar to the one based on jagged step functions in terms of shape and orientation, although the volume varied noticeably with the frequency limit [11, 10]. These statistical methods generally produced 95%-boundary ellipsoids that were similar in shape, orientation, and volume across different methods and basic collections, and were centered near the mean of the mismatch points [10, 8]. However, they were incapable of determining the “optimal boundary” containing all possible metamers [10].

Linear Programming Approaches

The goal of finding the optimal boundary of MMBs led to the application of linear programming (LP) techniques [29, 22]. Allen was the first to demonstrate the application of linear programming to this problem, calculating the theoretical limits of metamerism [29, 22]. In 1975, Ohta and Wyszecki extended Allen’s method, providing an explicit LP formulation [22, 15]. This technique allowed for the calculation of the optimal boundary within which all metamers must lie, showing that this boundary is generally much larger than the 95% statistical ellipsoids and is not elliptical in shape [22, 15]. The LP method involves formulating constraints based on color formation equations and physical limits on reflectance (0 to 1) and then optimizing (minimizing or maximizing) a linear functional representing a direction in color space to find points on the boundary [22, 23]. Their calculations showed that for spectrally sampled data, the extreme spectral reflectance functions defining the boundary typically had up to five wavelengths with intermediate reflectance values (between 0 and 1), with the rest being 0 or 1 [22]. They expected this to correspond to not more than five transitions in the theoretical continuous case [22, 15].

Ohta and Wyszecki also used the LP technique to estimate extreme errors in colorimetric calculations due to wavelength sampling or integration methods and to study the colorimetric significance of mercury emission lines [27, 26].

Simple Element Method

Schmitt proposed the “simple-element method” for systematically evaluating color metamer ensembles, an approach related to the description of object gamuts as multidimensional polyhedrons. [7, 2, 23, 31]. This method is based on the property that any metameric function is a linear combination of a set of “simple elements” [7] which are related to the apexes of the MMB, a convex polyhedron whose apexes are projections of simple elements [7]. This method aimed to give the limits of the MMB directly with less computation [7]. For reflectance problems under a change of illuminant, in a finite-element model, the simple elements corresponding to the MMB boundaries are expected to have only 3 transitions (wavelengths where reflectance changes between 0, 1, or intermediate values) [7]. This contrasted with Ohta and Wyszecki’s finding of up to 5 intermediate values [22, 27], which Schmitt explained was due to Ohta and Wyszecki seeking limits along lines in color space, not necessarily the vertices (apexes) [7]. Schmitt’s method was found suitable for analyzing observer differences and efficiently determining MMB limits [7].

Practical Evaluation using Computer Formulation

In 1988, Berns quantified illuminant metamerism for various colorant systems (acrylic-enamel, ABS polymer, polyester, nylon textiles) using computer colorant formulation based on Kubelka-Munk theory [5]. Metamers formulated to match under D_{65} were evaluated under test illuminants (A, F2, F11), forming empirical mismatch gamuts. Multivariate statistics were used to quantify the size, shape, and location of these empirical gamuts, assuming normality. They found that the empirical gamuts were statistically different for different illuminants and different material substrates [5]. Comparing their results to theoretical gamuts calculated without a smoothness constraint, similar those from LP methods, they found the empirical gamuts were about one third the size, suggesting that the complexity of real colorant systems and their limitations in producing spectrally extreme reflectances restrict the practical extent of metamer mismatching compared to the theoretical maximum [5].

Modern Boundary Computation Methods

Following earlier statistical and linear programming methods, and the metamer set approach, recent advancements in MMB computation have focused on precisely characterizing the boundary of the metamer mismatch volume, overcoming previous limitations by not making restrictive assumptions about the shape of reflectance spectra or illuminant properties. The following sections review the key modern approaches that define the current state-of-the-art in MMB computation.

Finlayson and Morovic’s Metamer Set Algorithm

Finlayson and Morovic introduced a method to solve for a set of physically realizable and natural reflectance spectra, or “metamer set”, that map to the same color signal for a given color mechanism [2, 32]. Their approach is based on linear models of reflectance spectra, such as those derived from Principal Component Analysis (PCA), and physical constraints (reflectance values between 0 and 1) [2, 32, 33]. They showed that the metamer set is a convex, infinite, and bounded set [2]. They demonstrated how to compute this set in practice using computational geometry tools, such as convex hull algorithms [2], defining the theoretical MMB as the linear image of this set under a different illuminant [2, 32]. Building upon metamer mismatching concepts, Zhang, Funt, and Mirzaei investigated the phenomenon “in practice versus theory” [24] by using a large dataset of over 25 million reflectance spectra to compute empirical MMBs [24]. For a given color signal under a first illuminant, they identified reflectances within their dataset that produced that color signal, computationally relit these specific reflectances with a second illuminant, and computed the convex hull of the resulting color signals to define the empirical body [24]. Comparing these empirical bodies to theoretical ones computed using the algorithm of Logvinenko et al. [24, 3], they found that the empirical volumes are substantially smaller than theoretical ones (sometimes having a diameter about 15% of the theoretical diameter) but still show a systematic variation in size with Munsell value and chroma [24]. They also showed that properties of the empirical MMB, such as its centroid, could be used to make better predictions of color signals under illuminant changes than state-of-the-art methods like CAT02 [24].

Logvinenko et al.'s Algorithm

In 2014, Logvinenko, Funt, and Godau aimed to overcome limitations of methods to solve for a set of physically realizable and natural reflectance spectra by precisely characterizing the theoretically exact MMB boundary for any possible reflectance spectrum, moving beyond earlier restrictions [3]. They related the MMB boundary to a kind of six-dimensional object-color solid [3]. Building on theoretical work suggesting that the boundary of such volumes is defined by spectral reflectance functions corresponding to zero-crossings of linear combinations of sensor sensitivities, they proved that while the number of transitions in these functions could be large, the boundary of the 6-dimensional volume, although not limited to 5 transitions, could be parameterized by 5 wavelengths [3, 13]. Nonetheless, to simplify computation, they used a 5-transition approximation, generating points, some of which will lie slightly inside the true MMB [3]. However, a subsequent theoretical analysis has suggested that, unlike optimal spectra for single object-color solids for human vision, which have at most 4 transitions [14], that double object-color solids may have significantly more transitions for the optimal spectra that define its boundary. [16] Work on computing the precise boundary for metamer mismatch bodies [3, 34] (which this relates to) demonstrated that theoretical mismatch bodies can be larger than expected, but the authors acknowledged that the accuracy of this 5-transition approximation relative to the true volume remained unknown [3, 34]. The algorithm was implemented in MATLAB and involved optimization [3].

Centore's Algorithm

Centore presented an open-source algorithm for constructing MMBs using a linear programming approach in 2017 [23, 35]. While acknowledging the history of LP in this problem, the goal was to provide a simple, accessible, and rigorous method with readily available code [23]. The algorithm views the MMB as the convex hull of its vertices [23, 35]. It finds vertices by formulating and solving an LP problem that minimizes or maximizes a linear functional in a given direction, subject to the constraints defining the metamer set (color formation equation and 0–1 reflectance bounds) [23, 35]. Unlike the 5-transition approximation, this LP method does not inherently limit the number of transitions in the resulting optimal spectra [23, 35].

Stiebel and Merhof's Algorithm

Building on the theoretical framework by Logvinenko et al. [3], Stiebel and Merhof proposed a robust algorithm for computing MMB boundary points in 2018 [25]. Their method aimed to solve the original equations describing the precise boundary more reliably and efficiently [25]. The authors propose that a key advantage is that it suspends the requirement for illuminants to be strictly positive [25]. While previous theories suggest this is not a physically possible scenario, limits of metrology require algorithmic implementations of the theory to deal with such circumstances. [3, 24]. Stiebel and Merhof achieved this by describing the boundary through inequality constraints transferred from a dual boundary construction, relating to hyperplanes tangent to the object color solid [25]. The algorithm was generalized for arbitrary dimensionality (e.g., multispectral systems) [25]. They validated it against Logvinenko's theory for strictly positive illuminants [25] and suggest that the five-transition approximation may underestimated the true MMB volume, with many approximated points lying inside the true boundary [25].

Mackiewicz et al.'s Algorithms

In 2019, Mackiewicz, Rivertz, and Finlayson introduced two novel algorithms for calculating the theoretically maximal metamer mismatch bodies using spherical sampling [20]. Their methods calculate the precise maximum extent without imposing constraints on the number of transitions in the optimal spectra [20]. The two methods are based on: 1) linear programming optimization and 2) computational geometry using half-space intersection [20]. The LP method extremizes spherically sampled directions in what is known as “k-space” subject to constraints [14, 20]. The half-space intersection method computes the 6D object color solid and then calculates the 3D MMB as a cross-section [20]. Unfortunately, uniform sampling in k-space does not lead to uniform sampling in sensor (e.g., LMS) space [14]. Nonetheless, they found that the Logvinenko et al. (2014) five-transition approximation underestimated the true MMB volume somewhat, especially for illuminant changes involving less smooth spectra like F11 [20, 3]. Their analysis showed that the majority of optimal spectra on the true MMB boundary have more than five transitions [20]. The half-space intersection method was found to be significantly faster, calculating accurate volumes in seconds or less for sufficient samples [20].

Kuan Xu et al.'s Algorithm

In 2023, Kuan Xu, Ma, and Li [36] introduced a high-dimensional spherical sampling method, offering it as an enhancement to address limitations in prior spherical sampling approaches for MMB calculation. They noted that earlier methods, which built upon foundational work by Finlayson et al. [2, 37, 38] and Centore [23, 39], suffered from long computation times, a high incidence of duplicated boundary points, insufficient effective vertices, and a calculated metamer set dimension that often appeared lower than its theoretical counterpart [36]. Their method directly samples the metamer set to identify boundary points by systematically selecting “direction vectors” in the 31-dimensional spectral reflectance space and “polarizing”, or finding maxima and minima along these directions [36]. For each chosen direction vector, the algorithm leverages linear programming to maximize or minimize the projection of the spectral reflectance function onto that vector [36]. This optimization is performed under constraints that include satisfying the color formation equation for a target color and ensuring that reflectance values remain within physically realizable limits (0 to 1) [36]. Direction vectors are systematically constructed, initially setting single elements to 1 (or -1) and others to 0, then extending to combinations involving two and three dimensions to achieve a more comprehensive sampling of the metamer set’s boundary [36]. While the authors state that their method is “significantly faster” [36] than the “ordinary spherical sampling method” described by Centore [23], they note its performance is comparable to other modern linear programming techniques, such as the linear programming from Mackiewicz et al [20].

3 Applications of Metamer Mismatch Bodies

Metamer mismatch bodies (MMBs), which represent the range of potential color signals an object can produce under a second viewing condition given its color signal under a first condition [15, 3], have several practical applications in color science and imaging. A significant consequence of metamer mismatching is that it imposes a limit on color constancy, as a single color signal under one illuminant can correspond to an infinite convex set of color signals under a different illuminant [24]. This inherent ambiguity means that predicting how a color signal will change with an illumination change is uncertain [24, 6, 34]. Despite this, MMBs offer a way to approach this prediction problem. Research has shown that properties of the empirical MMB, such as its centroid, can be used to predict what a given color signal is likely to become under a specified illuminant, and this approach has been demonstrated to be a better predictor than some state-of-the-art methods like the CAT02 transform or the Gaussian metamer method [24]. The size of the MMB is directly related to the potential prediction error. That’s to say, larger volumes indicate greater uncertainty [24].

MMBs are also applied as a means to evaluate the color rendering properties of light sources. It is expected that lights that lead to smaller MMBs will generally yield better color rendering [3]. The concept of the theoretical chromaticity-mismatch limits of metamers, which describes the boundary of the region of possible colorimetric mismatches, has been used as an attractive way of specifying the color-rendering properties of light sources [22, 26]. A metamer mismatch index has been introduced, defined as the ratio of the MMB volume to the object-color solid volume, providing a quantitative measure of the degree of metamer mismatching for a given color signal and change in viewing conditions [3]. This index can be used as a measure of the color rendering of lights [4, 3, 40].

In color imaging and camera systems, MMBs are highly relevant. They are important in camera sensor design, where sensors that produce large mismatch bodies when changing from human cone sensitivity to camera sensitivity are undesirable [20, 20, 3]. The size of the MMB induced by differences between sensor sensitivities (like human cones and camera sensors) can potentially quantify the error introduced [3]. MMBs are also relevant to color calibration and the color correction problem in imaging pipelines [20, 3]. Metamer-set-based approaches, which calculate the entire set of metameric reflectances for a given color signal [2], utilize the resulting MMB as a representation of the uncertainty or error bars associated with reflectance estimation from camera responses [32, 2]. Studies have shown that these metamer-set-based methods can provide improved accuracy in reflectance estimation compared to traditional techniques [32]. MMBs or mismatch gamuts are also used to evaluate observer metameric mismatching [5, 8, 10] and color difference sensitivity [3, 41, 14].

MMBs also find practical applications in industrial color management and formulation. They have been used to quantify illuminant metamerism for specific coloration systems, such as textiles, paint, and polymer systems, by analyzing the resulting metameric mismatch gamuts [5, 5]. The extent of metamer mismatching has a direct bearing on assessing the color-rendering properties of light sources and the adequacy of daylight simulators for colorimetry [15]. The choice of the reference illuminant in colorant formulation significantly affects the size of the resulting metameric mismatch gamuts when evaluated under different test illuminants, influencing the success probability of formulating matches that remain relatively stable under a range of different illuminants [5].

4 An Optimized MATLAB implementation of Logvinenko, et. al.

To support the practical use of the Logvinenko et al. algorithm [21], a MATLAB-based implementation was developed by Godau as part of the original research effort. This code has since served as the de facto reference for experimental work relying on MMB boundary estimation using the five-transition approximation. However, as MMB analysis has expanded into areas requiring large-scale spectral evaluation, such as light source characterization, camera sensor design, and empirical validation of metamer mismatching, the computational limitations of the original implementation have become increasingly apparent.

Esposito et al. [42] directly criticized the computational performance of this codebase, reporting execution times of approximately 30 seconds per spectral power distribution for the metamer mismatch color rendering index (MMCRI) [43]. This made batch analysis infeasible and posed a barrier to incorporating MMB-based methods into larger imaging or lighting evaluation pipelines. To address the execution-time issue, the implementation has now been comprehensively re-engineered. Algorithmic optimizations and structural changes reduce typical execution times by more than two orders of magnitude for individual methods, enabling analyses involving thousands of spectra to be executed in practical time frames on standard hardware.

These performance gains are accompanied by improvements in modularity, readability, and ease of integration. The updated codebase features a more consistent and transparent API, making the logic easier to interpret and modify. Parameters are explicitly handled, documentation is embedded throughout, and the design facilitates adaptation to new research contexts for color mechanisms. These updates position the implementation as a more general-purpose tool for MMB exploration.

Finally, the new version incorporates automated test, coverage, and a containerized [44] MATLAB execution environment [45] to support reproducibility across systems and configurations. While the underlying algorithm remains rooted in the five-transition approximation proposed by Logvinenko et al., the resulting tool is significantly faster, easier to use, and more maintainable to serve as a robust foundation for future research into metamer mismatching and the application of Metamer Mismatch Bodies. The code has been made available via the Simon Fraser Computational Vision Laboratory GitHub [46] under the AGPL 3.0 License [47].

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